Test Series No.

1.10

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 14/03/2026

ANSWER KEY

1.	(1)	46.	(2)	91.	(3)	136.	(2)
2.	(2)	47.	(3)	92.	(2)	137.	(2)
3.	(2)	48.	(1)	93.	(3)	138.	(3)
4.	(3)	49.	(2)	94.	(3)	139.	(4)
5.	(2)	50.	(1)	95.	(1)	140.	(1)
6.	(2)	51.	(1)	96.	(1)	141.	(4)
7.	(2)	52.	(4)	97.	(2)	142.	(2)
8.	(2)	53.	(1)	98.	(3)	143.	(1)
9.	(3)	54.	(2)	99.	(3)	144.	(3)
10.	(1)	55.	(1)	100.	(2)	145.	(2)
11.	(1)	56.	(4)	101.	(3)	146.	(2)
12.	(3)	57.	(1)	102.	(2)	147.	(4)
13.	(3)	58.	(2)	103.	(3)	148.	(2)
14.	(3)	59.	(1)	104.	(3)	149.	(3)
15.	(2)	60.	(4)	105.	(2)	150.	(2)
16.	(3)	61.	(4)	106.	(1)	151.	(1)
17.	(2)	62.	(1)	107.	(1)	152.	(2)
18.	(2)	63.	(1)	108.	(4)	153.	(3)
19.	(4)	64.	(1)	109.	(4)	154.	(2)
20.	(2)	65.	(3)	110.	(3)	155.	(1)
21.	(2)	66.	(1)	111.	(2)	156.	(2)
22.	(1)	67.	(4)	112.	(1)	157.	(3)
23.	(4)	68.	(1)	113.	(3)	158.	(1)
24.	(3)	69.	(1)	114.	(1)	159.	(4)
25.	(4)	70.	(1)	115.	(2)	160.	(2)
26.	(1)	71.	(2)	116.	(3)	161.	(1)
27.	(2)	72.	(1)	117.	(2)	162.	(2)
28.	(2)	73.	(1)	118.	(3)	163.	(2)
29.	(3)	74.	(1)	119.	(3)	164.	(3)
30.	(3)	75.	(1)	120.	(3)	165.	(2)
31.	(2)	76.	(1)	121.	(2)	166.	(1)
32.	(3)	77.	(4)	122.	(3)	167.	(2)
33.	(1)	78.	(1)	123.	(2)	168.	(4)
34.	(3)	79.	(1)	124.	(1)	169.	(1)
35.	(3)	80.	(3)	125.	(1)	170.	(4)
36.	(2)	81.	(3)	126.	(1)	171.	(2)
37.	(3)	82.	(2)	127.	(3)	172.	(1)
38.	(4)	83.	(1)	128.	(4)	173.	(4)
39.	(2)	84.	(1)	129.	(2)	174.	(4)
40.	(1)	85.	(4)	130.	(1)	175.	(1)
41.	(1)	86.	(3)	131.	(4)	176.	(4)
42.	(3)	87.	(1)	132.	(1)	177.	(3)
43.	(3)	88.	(1)	133.	(3)	178.	(1)
44.	(2)	89.	(2)	134.	(4)	179.	(1)
45.	(1)	90.	(4)	135.	(2)	180.	(3)

[Contd...

Hints & Solutions

1. (1)

Planck's constant, $(h)=[M^1L^2T^{-1}]$

Speed of light, $(c)=[L^1T^{-1}]$

Universal gravitational constant

$(G)=[M^{-1}L^3T^{-2}]$

$$[f]=\frac{\sqrt{M^1L^2T^{-1}\times L^5\times T^{-5}}}{\sqrt{M^{-1}L^3T^{-2}}}=\left[M^1L^2T^{-2}\right]$$

(NCERT Ed. 2025-26, 11th, Page No. 7)

2. (2)

They collide when the surfaces touch. Total distance to cover = Separation – Sum of radii.

$S_{total}=9r-(r+2r)=6r$.

Since Center of Mass is stationary, $m_1x_1=m_2x_2$.

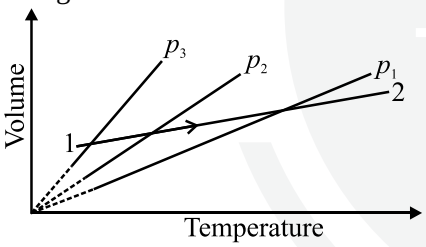
$m(x)=2m(6r-x)\Rightarrow x=12r-2x$

$\Rightarrow 3x=12r\Rightarrow x=4r$

(NCERT Ed. 2025-26, 11th, Page No. 99)

3. (2)

If this line had the tendency to pass through the origin, the pressure had been constant, hence answer (1) is wrong. Look at this process in the background of several isobaric lines.



Here note that $p_1 > p_2 > p_3$. The process line 1 - 2 cuts p_3 first then p_2 then p_1. So during the process 1 - 2 the pressure continuously increases.

(NCERT Ed. 2025-26, 11th, Page No. 246)

4. (3)

$\Delta D=D\alpha\Delta T$

$0.02=10.00\left(2\times10^{-5}\right)\Delta T$

$$\Delta T=\frac{0.02}{20\times10^{-5}}=\frac{2000}{20}=100^\circ\text{C}$$

$T_{final}=20+100=120^\circ\text{C}$

(NCERT Ed. 2025-26, 11th, Page No. 205)

5. (2)

Time period $T=2\pi\sqrt{\frac{m}{K_{eq}}}$.

T is maximum when K_{eq} is minimum.

(1), (3), (4) are parallel combinations: $K_{eq}=2K$.

(2) is a series combination:

$$\frac{1}{K_{eq}}=\frac{1}{K}+\frac{1}{K}\Rightarrow K_{eq}=\frac{K}{2}.$$

Since $\frac{K}{2}$ is the minimum equivalent stiffness, the time period is maximum in option (2).

(NCERT Ed. 2025-26, 11th, Page No. 267)

6. (2)

$I_S=\frac{2}{5}MR^2$ and $I_D=\frac{1}{2}MR^2$

Given $5\left(\frac{2}{5}MR^2\right)=x\left(\frac{1}{2}MR^2\right)$

$2MR^2=\frac{x}{2}MR^2\Rightarrow 2=\frac{x}{2}\Rightarrow x=4$

(NCERT Ed. 2025-26, 11th, Page No. 116)

7. (2)

Since missile is launched with speed less than escape velocity, sum of its kinetic and potential energy will be negative.

(NCERT Ed. 2025-26, 11th, Page No. 135)

8. (2)

Conservation of Angular Momentum: $L_i=L_f$.

$I_1\omega_1+I_2\omega_2=(I_1+I_2)\omega_f$

$2(6)+4(3)=(2+4)\omega_f\Rightarrow 12+12=6\omega_f$

$\Rightarrow 24=6\omega_f\Rightarrow\omega_f=4$ rad/s

$KE_f=\frac{1}{2}I_{total}\omega_f^2=\frac{1}{2}(6)(4)^2=3\times16=48$ J.

(NCERT Ed. 2025-26, 11th, Page No. 122)

9. (3)

This is a physical pendulum.

The formula for time period is $T=2\pi\sqrt{\frac{I}{mgd}}$

Moment of inertia about the end $I=\frac{ml^2}{3}$

Distance from pivot to center of mass $d=\frac{l}{2}$

$$T=2\pi\sqrt{\frac{ml^2/3}{mg(l/2)}}=2\pi\sqrt{\frac{2l}{3g}}$$

10. (1)

Initial KE = $\frac{1}{2}mv^2=\frac{1}{2}(2)(20)^2=400$ J

Work done = $\int_0^5-2x\,dx=[-x^2]_0^5=-25$ J

Final KE = Initial KE + Work = $400-25=375$ J

(NCERT Ed. 2025-26, 11th, Page No. 76)

[Contd...

11. (1)

$$\vec{I} = -\left[\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right]$$

$$\vec{I} = -\left[-2\hat{i} - 2\hat{j} - 2\hat{k}\right] = 2\left(\hat{i} + \hat{j} + \hat{k}\right) \text{N/kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 134)

12. (3)

For a body falling from rest, distances covered in equal consecutive time intervals are in the ratio of odd numbers: $1:3:5:7\ldots$.

(NCERT Ed. 2025-26, 11th, Page No. 20)

13. (3)

Segment 1 (Semicircle):

Mass $m_1 = \text{length} \times \lambda = \pi R\lambda$; Center of mass $y_1 = \frac{2R}{\pi}$.

Segment 2 (Diameter):

Mass $m_2 = \text{length} \times \lambda = 2R\lambda$;

Center of mass $y_2 = 0$.

Combined COM

$$Y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{(\pi R\lambda)\left(\frac{2R}{\pi}\right) + (2R\lambda)(0)}{\pi R\lambda + 2R\lambda}$$

$$= \frac{2R^2\lambda}{R\lambda(\pi + 2)} = \frac{2R}{\pi + 2}$$

The Assertion is the mathematically correct result, and the Reason provides the correct fundamental principle used to derive it.

(NCERT Ed. 2025-26, 11th, Page No. 97)

14. (3)

Initial angular velocity

$$\omega_0 = \frac{v}{r} = \frac{10}{0.5} = 20 \text{ rad/s}$$

Angular acceleration

$$\alpha = (\omega_f - \omega_0)/t = (0 - 20)/5 = -4 \text{ rad/s}^2$$

Torque $\tau = I|\alpha| = 1 \times 4 = 4$ Nm

(NCERT Ed. 2025-26, 11th, Page No. 120)

15. (2)

Using $T^2 \propto r^3$: $\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^3 = \left(\frac{4r}{r}\right)^3 = 64$

$\frac{T_2}{T_1} = \sqrt{64} = 8$.

Thus, $T_2 = 8T$.

(NCERT Ed. 2025-26, 11th, Page No. 129)

16. (3)

The support must hold the weight of both the crate and the chain.

Total Weight = Weight of Crate + Weight of Chain

$W_{total} = 500 + (10 \times 10) = 500 + 100 = 600$ N.

(NCERT Ed. 2025-26, 11th, Page No. 58)

17. (2)

From the ideal gas law, $PV = NkT$.

Density $\rho = \frac{\text{Total Mass}}{\text{Volume}} = \frac{N \cdot m}{V}$

Rearranging the gas law: $\frac{N}{V} = \frac{P}{kT}$

Therefore, $\rho = \frac{Pm}{kT}$

(NCERT Ed. 2025-26, 11th, Page No. 203)

18. (2)

Area $A = \pi r^2 = 3.14159\ldots \times (1.5)^2 \approx 7.068$.

The radius (1.5) has 2 significant figures, so the answer must be rounded to 2 significant figures.

Result: 7.1 m^2.

(NCERT Ed. 2025-26, 11th, Page No. 5)

19. (4)

Acceleration:

$$S = ut + \frac{1}{2}at^2 \Rightarrow 100 = 0 + \frac{1}{2}a(100) \Rightarrow a = 2 \text{ m/s}^2$$

Equation of motion: $F_{applied} - f_k = ma$

$$60 - \mu_k(100) = 10(2) \Rightarrow 60 - 20 = 100\mu_k \Rightarrow \mu_k = 0.4$$

(NCERT Ed. 2025-26, 11th, Page No. 60)

20. (2)

Process I is a vertical line $\rightarrow$ constant volume $\rightarrow$ Isochoric (III).

Process IV is a horizontal line $\rightarrow$ constant pressure $\rightarrow$ Isobaric (II).

Between curves II and III, curve II is steeper $\rightarrow$ Adiabatic (I), and curve III is less steep $\rightarrow$ Isothermal (IV).

(NCERT Ed. 2025-26, 11th, Page No. 234)

21. (2)

Energy after impact (E_f) allows it to rise to height h.

$E_f = mgh$

Energy before impact (E_i) was $mgh + \frac{1}{2}mv_0^2$

Given $E_f = 0.5E_i \Rightarrow mgh = 0.5\left(mgh + \frac{1}{2}mv_0^2\right)$

$$2gh = gh + \frac{1}{2}v_0^2 \Rightarrow gh = \frac{1}{2}v_0^2 \Rightarrow v_0 = \sqrt{2gh}$$

$$v_0 = \sqrt{2 \times 10 \times 5} = 10 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 84)

22. (1)

L.C. $= 1 \text{ MSD} - 1 \text{ VSD}$.

$20 \text{ VSD} = 19 \text{ MSD} \Rightarrow 1 \text{ VSD} = \frac{19}{20} \text{ MSD}$.

L.C. $= 1 \text{ MSD} - 0.95 \text{ MSD} = 0.05 \text{ MSD} = 0.05 \text{ mm}$

[Contd...

23. (4)

Intensity $I \propto \frac{1}{r^2}$.

Distance decreases by factor of 10 ($10\text{m} \to 1\text{m}$), so Intensity increases by factor of $10^2 = 100$.

Change in dB $= 10\log(100) = 10(2) = 20$ dB.

New level $= 40 + 20 = 60$ dB.

24. (3)

In an $x-t$ graph, slope represents velocity.

Since $v_B > v_A$, the line for B must have a steeper slope than A.

Graph (3) correctly depicts B with a steeper slope catching up to and overtaking A.

(NCERT Ed. 2025-26, 11th, Page No. 23)

25. (4)

$F_x = F\cos\theta = 10\cos 60° = 5$

$F_y = F\sin\theta = 10\sin 60° = 5\sqrt{3}$

(NCERT Ed. 2025-26, 11th, Page No. 32)

26. (1)

Distance = Sum of areas under the velocity-time curve.

$0-1\text{s}: v$ reaches $at = 2$ m/s, $s_1 = \frac{1}{2}at^2 = 1$ m.

$1-2\text{s}: a = 0, v$ constant at 2 m/s, $s_2 = vt = 2\times 1 = 2$ m

$2-3\text{s}: v$ goes from 2 to 0 ($a = -2$),

$s_3 = ut + \frac{1}{2}at^2 = 2(1) - 1 = 1$ m.

Total distance $= 1 + 2 + 1 = 4$ m.

(NCERT Ed. 2025-26, 11th, Page No. 17)

27. (2)

Wave number $k = \frac{\pi}{4}$. Since $k = \frac{2\pi}{\lambda}$,

we get $\frac{\pi}{4} = \frac{2\pi}{\lambda} \Rightarrow \lambda = 8$ m.

Distance between consecutive nodes

$= \frac{\lambda}{2} = \frac{8}{2} = 4$ m.

(NCERT Ed. 2025-26, 11th, Page No. 290)

28. (2)

$u_x = 50\cos 53^\circ = 30$, $u_y = 50\sin 53^\circ = 40$

v_y at $t = 2$ s: $v_y = 40 - 10(2) = 20$

v_y at $t = 6$ s: $v_y = 40 - 10(6) = -20$

Average vertical velocity $= (20 + (-20))/2 = 0$

Average horizontal velocity $= 30$ (constant)

Net Avg Velocity $= \sqrt{30^2 + 0^2} = 30$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 38)

29. (3)

Change in momentum (Δp) = Area under $F-t$ graph.

$\Delta p = (\text{Area of Triangle } 0-2\text{s}) + (\text{Area of Rectangle } 2-4\text{s}) + (\text{Area of Rectangle } 4-8\text{s})$

$\Delta p = \left(\frac{1}{2}\times 2\times 6\right) + (2\times -3) + (4\times 3)$

$= 6 - 6 + 12 = 12$ Ns

(NCERT Ed. 2025-26, 11th, Page No. 55)

30. (3)

Fundamental of closed pipe: $f_c = \frac{v}{4L_c}$.

First overtone (2nd harmonic) of open pipe:

$f_o = \frac{2v}{2L_o} = \frac{v}{L_o}$

Given $f_c = f_o \Rightarrow \frac{v}{4L_c} = \frac{v}{L_o} \Rightarrow L_o = 4L_c$

$L_o = 4\times 20$ cm $= 80$ cm

(NCERT Ed. 2025-26, 11th, Page No. 292)

31. (2)

Pressure $P = h\rho g = 1000\times 10^3 \times 10 = 10^7$ Pa.

Fractional compression $\frac{\Delta V}{V} = \text{Compressibility} \times P$

$\frac{\Delta V}{V} = (5\times 10^{-10})\times 10^7 = 5\times 10^{-3} = 0.5\times 10^{-2}$

(NCERT Ed. 2025-26, 11th, Page No. 173)

32. (3)

Internal forces occur in equal and opposite pairs, so they cannot change the net linear momentum of a system. However, they can do work (e.g., an internal explosion), which alters the total kinetic energy.

(NCERT Ed. 2025-26, 11th, Page No. 99)

33. (1)

$\vec{v}_A = 10\hat{i}$, $\vec{v}_B = 10\hat{j}$

$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = 10\hat{j} - 10\hat{i}$

Magnitude $= \sqrt{10^2 + (-10)^2} = 10\sqrt{2}$ m/s

Direction is North-West (Negative *x*, Positive *y*).

(NCERT Ed. 2025-26, 11th, Page No. 37)

34. (3)

Isothermal $V \to \frac{V}{2}$:

$P_1V_1 = P_2V_2 \Rightarrow P(V) = P_2(V/2) \Rightarrow P_2 = 2P$

Adiabatic $\frac{V}{2} \to \frac{V}{16}$: $P_2V_2^\gamma = P_3V_3^\gamma$.

[Contd...

$$2P\left(\frac{V}{2}\right)^{5/3} = P_3\left(\frac{V}{16}\right)^{5/3}$$

$$P_3 = 2P\left(\frac{V/2}{V/16}\right)^{5/3} = 2P(8)^{5/3} = 2P(2^3)^{5/3}$$

$$= 2P\left(2^5\right) = 2P(32) = 64P$$

(NCERT Ed. 2025-26, 11th, Page No. 235)

35. (3)

Volumetric strain $\frac{\Delta V}{V} = \epsilon(1-2\sigma)$.

If $\sigma = 0.5$, then $(1-2(0.5)) = 0$.

The volume remains constant.

(NCERT Ed. 2025-26, 11th, Page No. 174)

36. (2)

At critical condition, velocity at top C is $v_c = \sqrt{gR}$

Using conservation of energy from C to P (taking center as reference $h = 0$):

$$\frac{1}{2}mv_c^2 + mg(R) = \frac{1}{2}mv_p^2 + mg(R\cos 60°)$$

$$\frac{1}{2}gR + gR = \frac{1}{2}v_p^2 + \frac{1}{2}gR$$

$$\Rightarrow v_p^2 = 2gR \Rightarrow v_p = \sqrt{2gR}$$

(NCERT Ed. 2025-26, 11th, Page No. 79)

37. (3)

For collision, horizontal velocity components must match.

$$v_{proj}\cos\theta = v_{drone}$$

$$200\cos\theta = 100 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$

(NCERT Ed. 2025-26, 11th, Page No. 38)

38. (4)

According to Stokes' Law, the viscous drag force is given by $F = 6\pi\eta rv$. It directly depends on the viscosity coefficient (η), the radius/size of the body (r), and its velocity (v).

(NCERT Ed. 2025-26, 11th, Page No. 192)

39. (2)

By continuity equation,

$$A_{in}v_{in} = A_{out}v_{out}$$

$$\pi R^2 v = n(\pi r^2)v_{new}$$

$$v_{new} = \frac{vR^2}{nr^2}$$

(NCERT Ed. 2025-26, 11th, Page No. 187)

40. (1)

Standard equation: $\frac{d^2x}{dt^2} + \omega^2 x = 0$

Given: $\frac{d^2x}{dt^2} + 25x = 0$

Comparing, $\omega^2 = 25 \Rightarrow \omega = 5$

Time Period $T = \frac{2\pi}{\omega} = \frac{2\pi}{5}$ s

(NCERT Ed. 2025-26, 11th, Page No. 267)

41. (1)

Heat lost by water

$= ms\Delta T = 50\times 1\times(20-0) = 1000$ cal.

Heat gained by ice to melt mass m: $Q = mL$

$$1000 = m(80) \Rightarrow m = \frac{1000}{80} = 12.5 \text{ g}$$

(NCERT Ed. 2025-26, 11th, Page No. 213)

42. (3)

In both cases, the tension (T) in the wire is equal to W. Since the length (L), area of cross-section, and material (Young's modulus) are identical, the elongation $\Delta L = \frac{TL}{AY}$ remains the same.

(NCERT Ed. 2025-26, 11th, Page No. 170)

43. (3)

Excess Pressure $\Delta P = \frac{2T}{R}$

$$\Delta P = \frac{2\times 0.07}{2\times 10^{-3}} = \frac{0.14}{0.002} = 70 \text{ N/m}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 196)

44. (2)

Average translational kinetic energy per molecule depends only on temperature, given by $E = \frac{3}{2}kT$.

It is independent of the mass or nature of the gas. Thus, the ratio is 1:1.

(NCERT Ed. 2025-26, 11th, Page No. 250)

45. (1)

Volume $V = AL$ is constant.

$A = \pi r^2$

$$r' = \frac{r}{2} \Rightarrow A' = \frac{A}{4}$$

Since $A'L' = AL \Rightarrow \left(\frac{A}{4}\right)L' = AL \Rightarrow L' = 4L$

Heat rate $H \propto \frac{A}{L}$

$$H_{new} \propto \frac{A/4}{4L} = \frac{1}{16}\frac{A}{L} = \frac{Q}{16}$$

(NCERT Ed. 2025-26, 11th, Page No. 214)

[Contd...

46. (2)

The overall reaction A $\rightleftharpoons$ D is obtained by the adding the three equlibria:

A $\rightleftharpoons$ B, B $\rightleftharpoons$ C and C $\rightleftharpoons$ D.

When chemical equations are added, their equilibrium constants are multiplied. Therefore, the equilibrium constant for the overall reaction is:

$K = K_1 \times K_2 \times K_3 = 2.0 \times 3.0 \times 2.0 = 12.0$

Hence, the equlibrium constant for A $\rightleftharpoons$ D is 12.0.

[NCERT Class 11th Page No. 181]

47. (3)

Compound p is anti-aromatic.

Compound q is aromatic.

Compound r is non-aromatic.

The order of stability is: aromatic > non-aromatic > anti-aromatic.

Hence, q > r > p

[NCERT Class 11th Page No. 322]

48. (1)

False – Maximum kinetic energy depends on frequency (ν), not on intensity. Intensity only affects number of electrons.

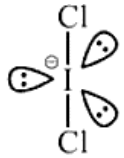
True – Above threshold frequency, increasing intensity increases the number of emitted photoelectrons (more photons → more electrons).

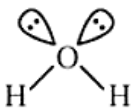
True – Threshold frequency (ν_0) exists; below it, no emission occurs regardless of intensity.

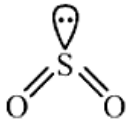
False – Stopping potential depends on maximum kinetic energy, hence depends on frequency, not independent of it.

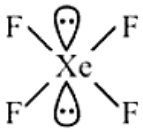
[NCERT Class 11th Page No. 40]

49. (2)

(A)  bond pair to lone pair ratio = 2 : 3

(B)  bond pair to lone pair ratio = 2 : 2

(C)  bond pair to lone pair ratio = 4 : 1

(D)  bond pair to lone pair ratio = 4 : 2

[NCERT Class 11th Page No. 115]

50. (1)

$$Ca + 2HCl \rightarrow CaCl_2 + H_2$$

1 mole of Ca produces 1 mole of H_2.

At STP. 1 mole gas = 22.4 L = 22400 mL

Given volume of H_2 = 112 mL

$$\text{Moles of } H_2 = \frac{112}{22400} = \frac{1}{200} = 0.005\text{mol}$$

From stoichiometry:

Moles of Ca = Moles of H_2 = 0.005 mol

Mass of required:

Mass = n × M = 0.005 × 40 = 0.20 g

[NCERT Class 11th Page No. 19]

51. (1)

Newlands' law of octaves states that elements were arranged in the increasing order of their atomic masses and every eighth element showed similar properties.

Mendeleev arranged elements mainly in the order of increasing atomic mass and observed that chemical properties of elements repeat periodically, which led to the formulation of the periodic law based on atomic mass.

[NCERT Class 11th Page No. 75]

52. (4)

$$(10L, 300K) \xrightarrow{n=1} (20L, 300K)$$

$$w = -nRT \ln \frac{V_2}{V_1}$$

Substituting the values, we have

$$= -1 \times 8.3 \times 300 \times \ln\left(\frac{20}{10}\right)$$

On solving, we get, we

w = –1.718 kJ

q = –w = 1.718 kJ

$\Delta U = q + w = 0$

[NCERT Class 11th Page No. 142]

53. (1)

In cold alkaline medium, bromine undergoes disproportionation to form bromide and hypobromite:

$Br_2 + 2OH^- \rightarrow Br^- + BrO^- + H_2O$

Here, one Br atom is reduced (0 → −1) to Br^- and the other is oxidized (0→+1) to BrO^-.

Comparing with the given general equation:

$aBr_2 + bOH^- \rightarrow cBrO^- + dBr^- + eH_2O$

From the balanced equation:

a = 1, b = 2, c = 1, d = 1

Thus, the correct set of coefficients is 1, 2, 1 and 1.

[NCERT Class 11th Page No. 250]

[Contd...

54. (2)

(a) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2$

$$[H^+] = \frac{20\times 20}{800} = \frac{1}{20} \Rightarrow pH = -\log\left(\frac{1}{20}\right)$$

$\therefore$ pH = 1.3

(b) $$\log\left(\frac{Kw_2}{Kw_1}\right) = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

(c) $Ka = 10^{-5}$, $pH = 5 \Rightarrow [H^+] = 10^{-5}$

$$Ka = \frac{c\alpha^2}{(1-\alpha)} \Rightarrow Ka = \frac{[H^+].\alpha}{(1-\alpha)}$$

$$10^{-5} = \frac{10^{-5}\times\alpha}{(1-\alpha)} \text{ or } \alpha = 50\%$$

[NCERT Class 11th Page No. 199]

55. (1)

(1) $[Fe(CN)_5NOS]^{4-} \rightarrow$ Violet colour

(2) $[Fe(SCN)]^{2+} \rightarrow$ Blood red colour

(3) $Fe_4[Fe(CN)_6]_3.H_2O \rightarrow$ Prussian blue colour

(4) $[Fe(CN)_6]^4 \rightarrow$ Yellow colour

[NCERT Class 11th Page No. 326]

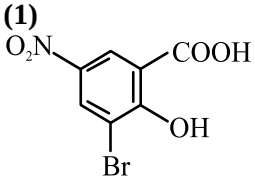
56. (4)

Most Group 14 tetrahalides (MX_4) are covalent, including $SnCl_4$. In fact, $SnCl_4$ is a covalent liquid, not ionic. Also, exceptions like SnF_4 and PbF_4 are more ionic in character due to high lattice energy and strong M–F interactions.

Fluorine, being highly electronegative and small in size, forms very strong and polar bonds, which can increase ionic character in fluorides compared to other halides.

[NCERT Class 11th Page No. 320]

57. (1)



The correct IUPAC name of the given compound is 3-Bromo-2-hydroxy-5-nitro-Benzoic acid.

[NCERT Class 11th Page No. 268]

58. (2)

(1) Electronic configuration (for $Z \le 7$):

$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\pi_{2p_z}^1\pi_{2p_y}^1$

Bond order = $\frac{(6-4)}{2} = 1$ and it has two unpaired electrons $\rightarrow$ paramagnetic.

(2) C_2:

Configuration:

$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{1s}^{*2}\pi_{2p_z}^2\pi_{2p_y}^2$

Bond order = $\frac{(8-4)}{2} = 2$ and all electrons are paired $\rightarrow$ diamagnetic.

(3) O_2:

Configuration includes π_{2p}^* electrons with two unpaired electrons.

Bond order = 2, but it is paramagnetic.

(4) O_2:

Extra electrons fill antibonding orbitals.

Bond order = $\frac{(9-6)}{2} = 1.5$ and it is paramagnetic.

[NCERT Class 11th Page No. 130]

59. (1)

A. Propanal and Acetone $\rightarrow$ These have the same molecular formula (C_3H_6O) but different functional groups (aldehyde vs ketone), so they show functional isomerism

B. Pentane and 2-methylbutane $\rightarrow$ They differ in carbon chain arrangement (straight chain vs branched chain) while having the same molecular formula (C_5H_{12}), so they exhibit chain isomerism

C. 1-Chloropropane and 2-Chloropropane $\rightarrow$ The functional group (C_l) is attached at different positions on the same carbon skeleton, hence position isomerism

D. Ethoxypropane and Methoxybutane $\rightarrow$ Both are ethers with the same molecular formula but different alkyl groups on either side of the oxygen atom, showing metamerism

[NCERT Class 11th Page No. 271]

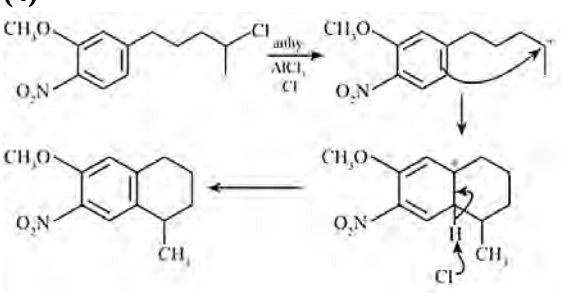
60. (4)

Spectral lines are produced in both absorption (lower $\rightarrow$ higher) and emission (higher $\rightarrow$ lower) transitions, not only when an electron absorbs energy.

Emission spectra are formed when electrons fall from higher to lower energy levels and emit photons of specific frequencies.

[NCERT Class 11th Page No. 43]

61. (4)



[NCERT Class 11th Page No. 323]

[Contd...

62. (1)

Hydronium ion (H_3O^+) in aqueous solution does not remain isolated and gets further solvated by water molecules.

Hydrated species such as $H_5O_2^+$ and $H_9O_4^+$ formed due to strong hydrogen bonding in aqueous medium.

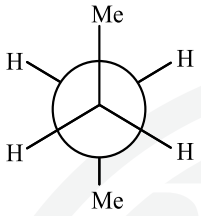
Hydration actually stabilizes the hydronium ion because solvation lowers its energy through ion–dipole interaction and hydrogen bonding.

Hydrated hydronium ions are stabilized by intermolecular hydrogen bonding with surrounding water molecules.

[NCERT Class 11th Page No. 188]

63. (1)

Conformation



Hence, it must be most stable.

due to two bulky methyl group are far apart from each other.

[NCERT Class 11th Page No. 307]

64. (1)

Work done = Area covered by the diagram

$\frac{1}{2}$ (Sum of parallel sides) × height

$$|w| = \frac{1}{2}(6+10)\times 6 = \frac{1}{2}\times 16\times 6 = 48\text{J}$$

[NCERT Class 11th Page No. 141]

65. (3)

We know the dissociation equation for a weak acid HA:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Since $[H^+] = Cx$ and $[A^-] = Cx$, while $[H A] = C(1-x)$, we get:

$$Ka = \frac{Cx^2}{1-x}$$

Taking log on both sides:

$\log Ka = \log C + 2\log x - \log(1-x)$

Since $pH = -\log[H^+] = \log(Cx)$, we get:

$pH = -\log C - \log x$

Similarly, $pKa = -\log Ka$, so;

$pKa = -\log C - 2\log x + \log(1-x)$

Now, subtracting:

$pH - pKa = (-\log C - \log x) - (-\log C)$

$2\log x + \log(1-x) = \log x - \log(1-x)$

$$= \log\frac{x}{1-x}$$

[NCERT Class 11th Page No. 201]

66. (1)

Sn^{4+}	69
Si^{4+}	40
Pb^{4+}	78
Ge^{4+}	53

[NCERT Class 11th Page No. 317]

67. (4)

Only (A) and (B) structure show resonance effect. In (A), benzyllic resonance is present while in (B), allylic resonance of carbocation takes place.

[NCERT Class 11th Page No. 276]

68. (1)

Addition of an electron to a neutral gaseous atom (electron gain enthalpy) is not always exothermic. Elements like noble gases and some alkaline earth metals have positive electron gain enthalpy (endothermic).

Removing an electron from an isolated gaseous atom always requires energy, so ionization enthalpy is always endothermic.

Nitrogen has a stable half-filled $2p^3$ configuration, while oxygen has one paired electron in 2p orbitals causing extra repulsion, making the first ionization enthalpy of oxygen slightly less than that of nitrogen.

Electronegativity is not constant; it depends on the chemical environment, oxidation state, and hybridization of the element in different compounds.

Electropositivity increases down Group 1, so Cs (and Fr) are more electropositive than Na; hence sodium is not the most electropositive alkali metal.

[NCERT Class 11th Page No. 89]

69. (1)

Let the fractional abundance of ^{24}X be x and that of ^{25}X be (1–x).

The average atomic mass is the weighted mean of isotopic masses:

$24x + 25(1-x) = 24.5$

$24x + 25 - 25x = 24.5 \Rightarrow 25 - x = 24.5 \Rightarrow x = 0.5$

Thus, fractional abundance of $^{24}X = 0.5$ and $^{25}X = 0.5$.

Therefore, the ratio $^{24}X : {}^{25}X = 0.5 : 0.5 = 1 : 1$.

[NCERT Class 11th Page No. 16]

[Contd...

70. (1)

$$2\left[\overset{+1}{Au}(CN)_2\right]^- + \overset{0}{Zn}(s) \rightarrow 2\overset{0}{Au} + \left[\overset{+2}{Zn}(CN)_4\right]^{-2}$$

It is a redox reaction in which Zn displaced Au^+

Reduction and oxidation both are taking place simultaneously.

[NCERT Class 11th Page No. 244]

71. (2)

Neopentane (2, 2-dimethylpropane) is highly symmetrical. All twelve hydrogen atoms in the molecule are equivalent because the three methyl groups are identical and arranged symmetrically around the central carbon. Hence, free radical chlorination at any hydrogen position gives the same monochloro product, resulting in only one monochloro derivative.

Neopentane is the most branched isomer of pentane. Increased branching reduces surface area and weakens van der Waals forces between molecules, leading to a lower boiling point compared to the straight-chain isomer n-pentane.

[NCERT Class 11th Page No. 303]

72. (1)

7-Hydroxyheptan-2-one is correct IUPAC name of given compound.

[NCERT Class 11th Page No. 268]

73. (1)

$1s^2\ 2s^2\ 2p^1 \rightarrow$ Boron (B)

$1s^2\ 2s^2\ 2p^2 \rightarrow$ Carbon (C)

$1s^2\ 2s^2\ 2p^4 \rightarrow$ Oxygen (O)

$1s^2\ 2s^2\ 2p^5 \rightarrow$ Fluorine (F)

All these elements belong to the second period. Electronegativity increase across a period from left to right due to increasing effective nuclear charge and decreasing atomic size.

Thus, electronegativity order:

B < C < O < F

Which corresponds to:

P < S < Q < R

[NCERT Class 11th Page No. 80]

74. (1)

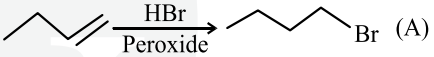
The dipole moment of NF_3 ($\approx$ 0.24 D) is smaller than that of NH_3 ($\approx$ 1.47 D) even though fluorine is more electronegative than hydrogen.

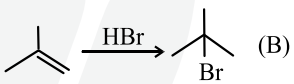
In trigonal pyramidal molecules, the lone pair on nitrogen contributes a dipole directed from N towards the lone pair region. In NH_3, the N–H bond dipoles point towards nitrogen and are in the same general direction as the lone pair dipole, so they reinforce each other, giving a larger net dipole moment. In NF_3, the highly electronegative fluorine atoms pull electron density strongly, so the N–F bond dipoles are directed away from nitrogen, opposite to the lone pair dipole. This results in partial cancellation of dipole vectors and hence a much smaller net dipole moment.

[NCERT Class 11th Page No. 112]

75. (1)

$$\text{Boiling point} \propto \frac{1}{\text{Branching}}$$

(A) 

(B) 

(C) 

[NCERT Class 11th Page No. 313]

76. (1)

In general for a weak electrolyte AxBy of concentration C, the degree of dissociation 'α' is given as,

$$\alpha = \left[\frac{K_{eq}}{(C)^{(x+y-1)}x^x.y^y}\right]^{\frac{1}{x+y}}$$

$$\underset{c(1-\alpha)}{A_2B_3(aq.)} \rightleftharpoons \underset{2c\alpha}{2A^{3+}_{(aq.)}} + \underset{3c\alpha}{3B^{2-}_{(aq)}}$$

$$Keq = \frac{[A^{3+}]^2[B^{2-}]^3}{[A_2B_3]} = \frac{4c^2\alpha^2 \times 27c^3\alpha^3}{c(1-\alpha)}$$

$$Keq = \frac{108c^5\alpha^5}{c} \quad \alpha = \left(\frac{Keq}{108c^4}\right)^{\frac{1}{5}}$$

[NCERT Class 11th Page No. 195]

[Contd...

77. (4)

Down the group, extent of back bonding decreases and tendency to act as Lewis acid increases.

$$\underset{(2p-2p)}{BF_3} > \underset{(2p-3p)}{BCl_3} > \underset{(2p-4p)}{BBr_3} > \underset{(2p-5p)}{BI_3}$$

(extent of back bonding)

$$BF_3 < BCl_3 < BBr_3 < BI_3$$

(lewis acidic nature)

[NCERT Class 11th Page No. 315]

78. (1)

The first ionisation enthalpy generally increases across a period because atomic size decreases and electrons are held more strongly.

Across a period, nuclear charge increases while electrons are added to the same shell, so shielding does not increase significantly, leading to an increase in effective nuclear charge.

The increased effective nuclear charge attracts valence electrons more strongly, making their removal more difficult and hence increasing the first ionisation enthalpy across the period.

[NCERT Class 11th Page No. 87]

79. (1)

According to the henderson hasselbalch equation,

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$$5.74 = 4.74 + \log\frac{[CH_3COONa]}{1}$$

$1 = \log[CH_3COONa]$

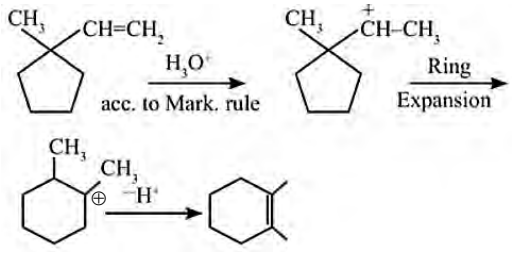
$[CH_3COONa] = 10\ M$

[NCERT Class 11th Page No. 201]

80. (3)

In this reaction, ring expansion i.e., 5-membered to

6-membered ring occurs to release ring strain.



[NCERT Class 11th Page No. 272]

81. (3)

For l = 3 (f subshell), number of orbitals

$= 2l + 1 = 2 \times 3 + 1 = 7.$

[NCERT Class 11th Page No. 55]

82. (2)

Given:

10 mL hydrocarbon produces 40 mL CO_2

So, 1 volume hydrocarbon produces $\frac{40}{10} = 4$ volumes of CO_2.

Hence, number of carbon atoms x = 4. The hydrocarbon is C_4H_y.

General combustion equation:

$$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$$

Given oxygen used = 60 mL for 10 mL hydrocarbon

Thus, 1 volume hydrocarbon requires $\frac{60}{10} = 6$ volumes of O_2.

So, $x + \frac{y}{4} = 6$

Substitute x = 4:

$$4 + \frac{y}{4} = 6 \Rightarrow \frac{y}{4} = 2 \Rightarrow y = 8$$

Therefore, molecular formula = = C_4H_8.

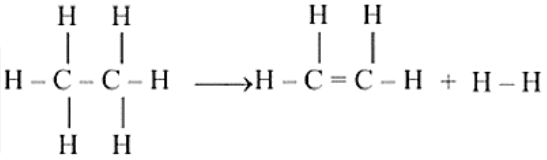
[NCERT Class 11th Page No. 19]

83. (1)

$C_2H_6 \rightarrow C_2H_4 + H_2$

When bond is formed, Energy is released

When bond is broken, Energy is consumed



$$\Delta H_f = [\Delta H_{C-C} + 6\Delta H_{C-H}] - \Delta H_{C=C} - 4\Delta_{C-H}$$

$$-\Delta H_{H-H} = \Delta H_{C-C} + 2\Delta H_{C-H} - \Delta H_{C=C} - \Delta H_{H-H}$$

Given: $\Delta H_{C=C} = 347 J/mol$

$\Delta H_{C=C} = 611 kJ/mol$

$\Delta H_{C-H} = 414 kJ/mol$

$\Delta H_{H-H} = 436 kJ/mol$

$= 347 + 2 \times 414 - 611 - 436 = 128$ kJ/mol

[NCERT Class 11th Page No. 154]

84. (1)

In N^{3-} ion 'N' is present in its lowest possible oxidation state, i.e., –3, therefore it cannot be reduced further because of which it cannot act as an oxidizing agent.

[NCERT Class 11th Page No. 242]

[Contd...

85. (4)

$$CH_3C \equiv CH \xrightarrow[H_2]{Pd/C}$$

$$CH_3 - CH = CH_2 \xrightarrow[(ii)Zn,H_2O]{(i)O_3}$$

$$CH_3CHO + HCHO$$

(B) (C)

[NCERT Class 11th Page No. 314]

86. (3)

$\Delta n_g = n_g$ (products) – n_g (reactants) = 2 – 2 = 0

Also, $K_p = \frac{(n_{HI})^2}{n_{H_2} n_{I_2}} \left(\frac{P_T}{n_T}\right)^{\Delta n_g}$

$n_{HI} = n_{H_2} = n_{I_2}$

$K_p = 1$

$1 = x \times 10^{-1}$

Hence, x = 10

[NCERT Class 11th Page No. 177]

87. (1)

The cis form of 2-butene is more polar than the trans form because in the cis isomer the substituents are on the same side of the double bond, leading to a net dipole moment.

In trans-2-butene, the two C–CH_3 bond dipoles are oriented in opposite directions due to the symmetrical arrangement across the double bond, so they cancel each other and the net dipole moment becomes zero.

Since the cancellation of dipole moments in the trans isomer explains why it is less polar than the cis isomer, the reason correctly explains the assertion.

[NCERT Class 11th Page No. 310]

88. (1)

(Nitro compounds and azo compounds do not give kjeldhal's test)

[NCERT Class 11th Page No. 289]

89. (2)

The most stable resonance structure is one which is neutral and has maximum number of double bonds.

$CH_3–CH=CH–\overset{\overset{O}{\|}}{CH}$ (III)

↓

More number of double bond

$CH_3–\overset{\oplus}{C}H–CH=CH-\overset{\ominus}{O}$ (II)

↓

Has –ve charge on more electronegative atom

$CH_3–\overset{\ominus}{C}H–CH=C(-\overset{\oplus}{O})–H$ (I)

↓

Has –ve charge on has less electronegative atom

Thus, stability order: III > II > I.

[NCERT Class 11th Page No. 276]

90. (4)

For emission line

$n_f < n_i$

$$\therefore \bar{v} = RZ^2\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right] = R\left[\frac{1}{8^2} - \frac{1}{n^2}\right]$$

or, $$\therefore \bar{v} = R_H\left(\frac{1}{64} - \frac{1}{n^2}\right) = \frac{R_H}{64} - \frac{R_H}{n^2}$$

$$\bar{v} = -R_H\left(\frac{1}{n^2}\right) + \frac{R_H}{64}$$

$\therefore$ y = mx + c

Slope = $-R_H$

[NCERT Class 11th Page No. 44]

[Contd...

91. (3)

Families are characterised on the basis of both vegetative and reproductive features of plant species.

(NCERT Ed. 2025-26, 11th, Page No. 5)

92. (2)

Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells.

(NCERT Ed. 2025-26, 11th, Page No. 88)

93. (3)

Auxin	Xylem differentiation
Cytokinin	Adventitious shoot growth
Gibberellins	Improves shape of apple fruit
Ethylene	Promotes rapid internode/petiole elongation in deep-water rice plant

(NCERT Ed. 2025-26, 11th, Page No. 176)

94. (3)

GA_3 (Gibberellic acid) → naturally occurring
Kinetin → does not occur naturally in plants
IBA (Indole-3-butyric acid) → naturally occurring
Zeatin → naturally occurring cytokinin
IBA → naturally occurring
ABA (Abscisic acid) → naturally occurring
2,4-D → synthetic auxin

(NCERT Ed. 2025-26, 11th, Page No. 175)

95. (1)

Most of the members of Chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts. Pyrenoids contain protein besides starch.

(NCERT Ed. 2025-26, 11th, Page No. 26)

96. (1)

Photosynthesis is under the influence of several factors, both internal (plant) and external. The plant factors include the number, size, age and orientation of leaves, mesophyll cells and chloroplasts, internal CO_2 concentration and the amount of chlorophyll. The external factors would include the availability of sunlight, temperature, CO_2 concentration and water.

(NCERT Ed. 2025-26, 11th, Page No. 149)

97. (2)

The lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner part. This ensures that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment.

(NCERT Ed. 2025-26, 11th, Page No. 93)

98. (3)

Chrysophytes includes diatoms and golden algae (desmids). They are found in fresh water as well as in marine environments. They are microscopic and float passively in water currents (plankton).

(NCERT Ed. 2025-26, 11th, Page No. 13)

99. (3)

In coconut, the mesocarp is fibrous.

(NCERT Ed. 2025-26, 11th, Page No. 66)

100. (2)

The stage between the two meiotic divisions is called interkinesis and is generally short lived. There is no replication of DNA during interkinesis.

(NCERT Ed. 2025-26, 11th, Page No. 127)

101. (3)

If the margins of sepals or petals overlap one another but not in any particular direction as in *Cassia* and gulmohur, the aestivation is called imbricate.
Cotton- Twisted aestivation
Pea- Vexillary aestivation
Calotropis – Valvate aestivation

(NCERT Ed. 2025-26, 11th, Page No. 64)

102. (2)

Plantae and Animalia kingdoms was developed that included all plants and animals respectively. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms.

(NCERT Ed. 2025-26, 11th, Page No. 10)

[Contd...

103. (3)

In bacteria, special membranous structure is the mesosome which is formed by the extensions of plasma membrane into the cell. Middle layer of cell envelope is occupied by cell wall.

(NCERT Ed. 2025-26, 11th, Page No. 90)

104. (3)

In pteridophytes, the spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus. These gametophytes require cool, damp, shady places to grow. Because of this specific restricted requirement and the need for water for fertilisation, the spread of living pteridophytes is limited and restricted to narrow geographical regions.

(NCERT Ed. 2025-26, 11th, Page No. 32)

105. (2)

In PS I the reaction centre chlorophyll a has an absorption peak at 700 nm, hence is called P700, while in PS II it has absorption maxima at 680 nm, and is called P680.

(NCERT Ed. 2025-26, 11th, Page No. 138)

106. (1)

Tracheids	Elongated, tapering ends, dead
Vessels	Cylindrical tube, multicellular, lignified
Sieve tube elements	Perforated end walls, lacks nucleus
Companion cells	Specialized parenchyma, helps in maintaining pressure gradient

(NCERT Ed. 2022-23, 11th, Page No. 87)

107. (1)

Heterocyst	Nitrogen fixation
Pellicle	Flexibility in Euglenoids
Flagellum	Motility in bacteria
Capsid	Protein coat of virus

(NCERT Ed. 2025-26, 11th, Page No. 13)

108. (4)

Human being is the only organism who is aware of himself, i.e., has self-consciousness.

(NCERT Ed. 2022-23, 11th, Page No. 5)

109. (4)

The pollen grain in gymnosperms is released from the microsporangium. They are carried in air currents and come in contact with the opening of the ovules borne on megasporophylls.

(NCERT Ed. 2025-26, 11th, Page No. 33)

110. (3)

In the photorespiratory pathway, there is neither synthesis of sugars, nor of ATP. Rather it results in the release of CO_2 with the utilisation of ATP. In the photorespiratory pathway there is no synthesis of ATP or NADPH.

(NCERT Ed. 2025-26, 11th, Page No. 147)

111. (2)

Agar, one of the commercial products obtained from *Gelidium* and *Gracilaria* are used to grow microbes and in preparations of ice-creams and jellies.

(NCERT Ed. 2025-26, 11th, Page No. 26)

112. (1)

The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata. During this phase the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation. By the end of diakinesis, the nucleolus disappears and the nuclear envelope also breaks down. Diakinesis represents transition to metaphase.

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113. (3)

Auxins (from Greek 'auxein' : to grow) was first isolated from human urine. Auxin promote apical dominance.

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114. (1)

When more than one carpel is present, they may be free (as in lotus and rose) and are called apocarpous. They are termed syncarpous when carpels are fused, as in mustard and tomato.

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[Contd...

115. (2)

In pteridophytes, the spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

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116. (3)

A monocot stem lacks phloem parenchyma. The vascular bundles in a monocot stem are conjoint and closed. The arrangement of vascular bundles in a ring-like manner is seen in a dicot stem. The presence of a large and well-developed pith is a feature of a monocot root.

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117. (2)

During electron transport in oxidative phosphorylation, oxygen acts as the final hydrogen acceptor.

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118. (3)

R.H. Whittaker (1969) proposed a Five Kingdom Classification. The kingdoms defined by him were named Monera, Protista, Fungi, Plantae and Animalia. The main criteria for classification used by him include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.

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119. (3)

The *cis* (forming) and *trans* (maturing) faces of Golgi apparatus are entirely different, although they are interconnected. Materials to be packaged (in the form of vesicles from the ER) fuse with the *cis* face and move toward the *trans* face, where they are eventually discharged. It was first observed by Camillo Golgi in 1898. George Palade is famous for discovering ribosomes in 1953.

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120. (3)

Marchantia	Gemma cup
Cycas	Coralloid roots
Fern	Prothallus
Sphagnum	Peat formation

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121. (2)

The cytoplasm is the main arena of cellular activities in both the plant and animal cells.

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122. (3)

The water splitting complex is associated with the PS II, which itself is physically located on the inner side of the membrane of the thylakoid.

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123. (2)

Asexual spores in ascomycetes are called conidia, which are produced exogenously on specialized mycelium known as conidiophores. Upon germination, the conidia produce mycelium.

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124. (1)

Growth, at a cellular level, is principally a consequence of increase in the amount of protoplasm. ABA plays an important role in seed development, maturation and dormancy. By inducing dormancy, ABA helps seeds to withstand desiccation and other factors unfavourable for growth.

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125. (1)

When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be actinomorphic, e.g., mustard, datura, chilli. When it can be divided into two similar halves only in one particular vertical plane, it is zygomorphic, e.g., pea, gulmohur, bean, *Cassia*. A flower is asymmetric (irregular) if it cannot be divided into two similar halves by any vertical plane passing through the centre, as in canna.

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126. (1)

During Anaphase I, the homologous chromosomes separate, while sister chromatids remain associated at their centromeres.

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127. (3)

All gymnosperms are heterosporous. Gymnosperms have albuminous cells and sieve cells. They lack sieve tubes and companion cells. Gymnosperms lack vessels in their xylem.

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128. (4)

Plants do not present great demands for gas exchange. Roots, stems and leaves respire at rates far lower than animals do.

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129. (2)

Endodermis in dicot root comprises a single layer of barrel-shaped cells without any intercellular spaces. The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.

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130. (1)

Solanum, *Petunia*, and *Datura* are three different genera. These three genera are all placed in the family Solanaceae. Plant families like Solanaceae and Convolvulaceae are included in the order Polymoniales.

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131. (4)

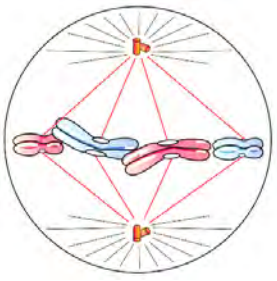


Figure represents metaphase of mitosis. The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis, hence the chromosomes are spread through the cytoplasm of the cell. During this stage, spindle fibres attach to kinetochores of chromosomes. At this stage, metaphase chromosome is made up of two sister chromatids, which are held together by the centromere

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132. (1)

Pneumatophores in *Rhizophora* are modified roots that grow vertically upwards to help get oxygen for respiration. The other three options—tendrils in cucumber, thorns in *Citrus*, and fleshy cylindrical structures in *Euphorbia*—are stem modifications.

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133. (3)

The F_1 headpiece is a peripheral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate. F_0 is an integral membrane protein complex that forms the channel through which protons cross the inner membrane. The passage of protons through the channel is coupled to the catalytic site of the F_1 component for the production of ATP.

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134. (4)

Natural classification systems developed, which were based on natural affinities among the organisms and consider, not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry. Such a classification for flowering plants was given by George Bentham and Joseph Dalton Hooker.

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135. (2)

The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER.

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136. (2)

In male cockroach, the sperms produced in the testes pass through the vas deferens and are stored in the seminal vesicles. Here they are later glued together to form spermatophores, which are discharged during copulation. The accessory gland secretions help in spermatophore formation, but storage of sperms is the primary function of seminal vesicles.

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137. (2)

Functional residual capacity = ERV + RV
= 1100 + 1200 = 2300 ml

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138. (3)

The tertiary structure gives an enzyme its specific three-dimensional shape, including the active site required for substrate binding and catalysis. If this structure is disrupted (denaturation), the active site changes shape, so the enzyme loses its biological activity even though the primary structure (peptide bonds) may remain intact.

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139. (4)

The passage of ova through the female reproductive tract is facilitated by the coordinated beating of cilia lining the oviduct (fallopian tube). These cilia help in moving the ovum towards the uterus.

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140. (1)

Hypothyroidism during pregnancy causes defective development and maturation of growing babies leading to stunted growth and mental retardation also called cretinism.

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141. (4)

Tendon and ligament are dense regular connective tissue. Cartilage is a specialised connective tissue.

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142. (2)

The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called the limbic lobe or limbic system.

Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions (e.g., excitement, pleasure, rage and fear), and motivation.

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143. (1)

Squamous epithelium is found in the walls of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary.

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144. (3)

Aldosterone is a hormone produced by the adrenal glands that regulates sodium reabsorption in the distal convoluted tubule (DCT) and collecting ducts of the kidneys. When aldosterone levels increase, sodium reabsorption is enhanced, leading to increased sodium retention by the kidneys.

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145. (2)

In a normal ECG, the QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction.

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146. (2)

Aplysia, *Aptenodyte* and *Anopheles* are triploblastic which contain three germ layers.

Aurelia is diploblastic which contains two germ layers.

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147. (4)

White fibers have less myoglobin than red fibers.
White fibers have less mitochondria than red fibers.
White fibers primarily use anaerobic respiration for energy.

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148. (2)

List-I	List-II
Platyhelminthes	Acoelomate and bilateral symmetry
Annelida	Longitudinal and circular muscles
Hemichordata	Indirect development and stomochord
Porifera	Ostia and spongocoel

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149. (3)

Petromyzon belongs to class Cyclostomata of sub phylum vertebrata and does not contain jaw, so it is a jawless vertebrate not invertebrate.

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150. (2)

The cerebellum has a very convoluted surface in order to provide the additional space for many more neurons.

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151. (1)

Secondary messengers are cyclic AMP, IP_3, Ca^{++} etc.

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152. (2)

List-I	List-II
Inspiratory capacity	Total volume of air a person can inspire after normal expiration
Expiratory capacity	Total volume of air a person can expire after a normal inspiration
Functional residual capacity	Volume of air that will remain in the lungs after a normal expiration
Vital capacity	The maximum volume of air a person can breathe in after a forced expiration

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153. (3)

Urine formed by the nephrons is ultimately carried to the urinary bladder where it is stored till a voluntary signal is given by the central nervous system (CNS). This signal is initiated by the stretching of the urinary bladder as it gets filled with urine. In response, the stretch receptors on the walls to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine. of the bladder send signals to the CNS. The CNS passes on motor messages

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154. (2)

Ionic gradients across the resting membrane are maintained by the active transport of ions by the sodium-potassium pump which transports 3 Na^+ outwards for 2 K^+ into the cell.

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155. (1)

List-I	List-II
Eosinophils	Resist infections and associated with allergic reaction
Basophils	Release histamine
Neutrophils	Phagocytosis
Lymphocytes	Immune response

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156. (2)

Venous blood returning from tissues has a higher $PCO_2 \approx 45$ mm Hg because tissues continuously produce CO_2.

Alveolar air has a lower $PCO_2 \approx 40$ mm Hg, so CO_2 diffuses from blood into the alveoli down its partial pressure gradient.

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157. (3)

Heavy meromyosin (HMM) forms the globular heads (cross arms) of the myosin filament. These heads have ATPase activity and binding sites for ATP and actin.

Troponin does not form the backbone; F-actin forms the backbone of the thin filament.

Tropomyosin helps cover the active sites on actin, but the masking in resting muscle is regulated by troponin, not tropomyosin independently.

Light meromyosin (LMM) mainly forms the tail of myosin and does not bind ATP or actin.

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158. (1)

Enzymes as catalysts differ from inorganic catalysts in many ways. Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures (say above 40°C).

Collagen is the most abundant protein in animal world and Ribulose bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

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159. (4)

Carbonic anhydrase is the enzyme which catalyses the formation of carbonic acid from carbon dioxide and water.

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[Contd...

160. (2)

Kidneys are reddish brown, bean shaped structures situated between the levels of the last thoracic and third lumbar vertebra close to the dorsal inner wall of the abdominal cavity.

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161. (1)

The adrenal cortex secretes many hormones, commonly called corticoids. The corticoids, which are involved in carbohydrate metabolism are called glucocorticoids. Glucocorticoids, particularly cortisol, produce antiinflammatory reactions and suppress the immune response.

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162. (2)

Asthma is a difficulty in breathing causing wheezing sound due to inflammation of bronchi and bronchioles.

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163. (2)

Oxytocin is synthesised in the hypothalamus (mainly in the paraventricular nuclei). It is then transported to the posterior pituitary, where it is stored and released into the blood when required, such as during uterine contractions and milk ejection.

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164. (3)

The association areas in the brain are primarily involved in processing and integrating information from different sensory modalities, facilitating communication, and supporting complex functions such as memory. However, the regulation of sexual behavior is more closely associated with specific areas of the brain, such as the hypothalamus and limbic system, rather than the association areas.

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165. (2)

Arthropods are triploblastic (three germ layers), bilaterally symmetrical, and possess an open circulatory system in which haemolymph directly bathes the tissues. They are also segmented animals with a chitinous exoskeleton and jointed appendages.

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166. (1)

Terrestrial amphibians are ureotelic whereas aquatic amphibians, aquatic insects and bony fishes are ammonotelic.

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167. (2)

According to the general principle of temperature effect on enzyme activity, the reaction rate halves for every 10°C decrease.

From 40°C to 30°C → rate becomes 8 → 4 units

From 30°C to 20°C → rate becomes 4 → 2 units

So, the approximate rate at 20°C = 2 units.

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168. (4)

The medulla oblongata does continue into the spinal cord through the foramen magnum.

The brain and spinal cord together form the central nervous system and are protected by cranium and vertebral column.

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169. (1)

Sternum is a flat bone on the ventral midline of thorax.

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170. (4)

Each artery and vein consists of three layers: an inner lining of squamous endothelium, the tunica intima, a middle layer of smooth muscle and elastic fibres, the tunica media, and an external layer of fibrous connective tissue with collagen fibres, the tunica externa. The tunica media is comparatively thin in the veins.

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171. (2)

The thorax of a cockroach consists of three segments: Prothorax, Mesothorax Metathorax

Each thoracic segment bears one pair of legs, and the mesothorax and metathorax also bear wings. Antennae are present on the head.

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172. (1)

The elemental composition in the human body in descending order is Oxygen (65%) > Carbon (18.5%) > Nitrogen (3.3%) > Calcium (1.5%).

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173. (4)

Cortisol and erythropoietin stimulate synthesis of RBC (erythropoiesis).

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[Contd...

174. (4)

Coronary Artery Disease, often referred to as atherosclerosis, affects the vessels that supply blood to the heart muscle. It is caused by deposits of calcium, fat, cholesterol and fibrous tissues, which makes the lumen of arteries narrower.

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175. (1)

Synovial joints are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.

Force generated by the muscles is used to carry out movement through joints, where the joint acts as a fulcrum.

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176. (4)

List-I	List-II
20-25% of CO_2	Transported by RBCs as carbamino-haemoglobin
100 ml of deoxygenated blood	Deliver around 4 ml of CO_2 to the alveoli
3% of O_2	Transported in dissolved state through the plasma
100 ml of oxygenated blood	Deliver around 5 ml of O_2 to the tissue

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177. (3)

Tetany is caused by low Ca^{2+} levels (hypocalcemia). Gout occurs due to accumulation of uric acid crystals. Muscular dystrophy is a genetic disorder (autoimmune disorder refers to Myasthenia gravis). Osteoporosis is an age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.

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178. (1)

List-I	List-II
Marine bony fish	Sea horse
Freshwater fish	Katla
Aquarium fish	Angel fish
Cartilaginous fish	Sting ray

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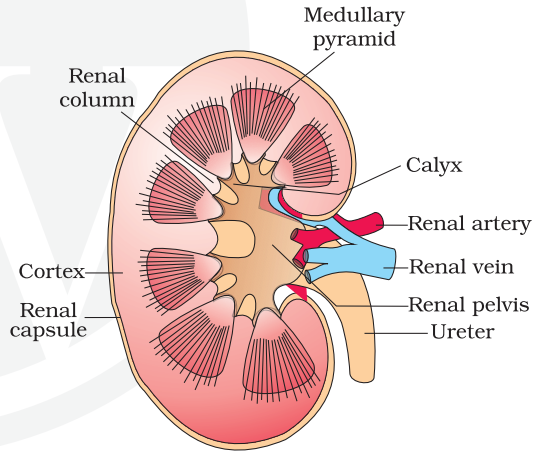
179. (1)

The SAN can generate the maximum number of action potentials, i.e., 70-75 min^{-1}, and is responsible for initiating and maintaining the rhythmic contractile activity of the heart. Therefore, it is called the pacemaker.

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180. (3)

'C' represents renal pelvis in the given figure.



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